Subject: Candidate Assessment — Real-time Attribution Dashboard (72h)

Hi — thanks for applying. This take-home mirrors work we do here.

Role: Data Engineer
Time: 72 hours
Submit to: jobs@customerlabs.co (include Git repo link, demo link, README)
Clarifying questions: Up to 5 questions within the first 24 hours (reply to this email)

Goal

Build a near-real-time attribution pipeline and dashboard that computes **First-Click** and **Last-Click** from a GA4 public dataset using **BigQuery + dbt**, plus a simple streaming demo.

Required deliverables

1. **Architecture (1 page):** diagram, tools, dataset/table names.
2. **dbt project:** stg_ → int_ /mart_ models, attribution models (First/Last), dbt tests, docs. State assumptions (lookback window, identity resolution, tie-breakers).
3. **Streaming demo:** script or pipeline that streams 5–20 sample events into BigQuery and demonstrates materialization. Explain dedupe/idempotency and expected latency.
4. **Realtime dashboard prototype:** First vs Last totals, 14-day time series, channel breakdown, and a live panel showing streamed events. (Looker Studio / Metabase / Streamlit / simple web app OK.)
5. **README & runbook:** run instructions, failure handling, monitoring suggestions, cost notes.
6. **Demo:** 5–8 minute screencast OR be ready to demo live.

Anti-AI & authenticity checks (required)

- Git commit history with incremental commits.
- worklog.md (6–10 short entries).
- At least two sketches/notes (photos OK).
- Be prepared for a 10–15 minute live walkthrough and a 15-minute live SQL edit during interview.

Evaluation (high level)

- dbt & modeling: 30 pts
- Attribution logic: 25 pts
- Streaming & realtime: 20 pts
- Dashboard UX & realtime: 15 pts
- Docs & ops readiness: 10 pts
  (≥85 excellent, 70–84 hireable, <70 revisit)

Practical notes

- Use any GA4 public dataset in BigQuery — state exact dataset/project. If you lack BigQuery access, provide runnable SQL + sample data and clear reproduction steps.
- Do not share production credentials. If hosting the dashboard isn't possible, provide a clear recording.

Suggested timeline

Day 0: clarify questions & pick dataset ✓
Day 1: staging models + tests; architecture doc ✓
Day 2: attribution models + streaming demo ✓
Day 3: dashboard, polish README, record demo, submit ✓

How to submit

Email jobs@customerlabs.co with: repo link (or zip), dashboard link or demo video, 1-page assumptions/edge cases, worklog.md, and sketches.

